

Midterm Study Guide**01- INTRO TO COGNITIVE PSYCHOLOGY**

1. What is Cognitive Psychology?
2. How do we study Cognition?
 - a. Experimental Psychology
 - b. Neuropsychology
3. Describe the early work in Cognitive Psychology
 - a. Donders
 - i. Simple RT
 - ii. Choice RT
 - b. Ebbinghaus
 - i. Forgetting curve
 - ii. savings
 - c. Wundt
 - i. Structuralism
 - ii. Analytic introspection
 - d. James
 - i. Firsts?
4. Describe the early work done by behaviorists which set the stage for the Cognitive Revolution.
 - a. Tolman
 - i. Cognitive Map
 - b. Other behaviorists- what did they think about studying cognition?
5. What is the “Cognitive Revolution” and how was it impacted by the advent of the computer?
 - a. Began in 1950s
 - b. Gradual
 - c. Information processing approach/models

02-INTRO TO COGNITIVE NEUROSCIENCE

1. What are the main parts of a neuron?
 - a. Soma
 - b. Dendrites
 - c. Axon
 - d. Other aspects?
2. How is information transferred in the nervous system?
 - a. Neural Firing
 - i. Neurons receive a signal that is in either a chemical form or a physical form.
 - ii. Sodium diffuses down through the dendrites and cell body.
 - iii. The electrical signal is activated and travels quickly down the axon to the axon terminals.

1. Steps in an Action Potential
 - a. Where?
 2. Myelination- why?
 3. All or None Law
 - iv. This causes the release of neurotransmitters at the synapse and the process starts all over in the next neuron.
3. How is information represented in the nervous system?
 - a. Neural Representation
 - i. Feature Detectors
 - ii. Stimulus Coding
 1. Specificity Coding
 2. Population Coding
 3. Sparse Coding
4. What are some of the ways we can study neural communication & what are the pros & cons of each?
 - a. Neuronal Recording
 - i. Single Cell Recording
 1. Receptive Fields
 2. Pros and Cons?
 - ii. EEG/ERPs
 1. Pros and Cons?
 - b. Spatial Localization
 - i. Single Cell Recording
 - ii. Neuropsychology
 - iii. MRI
 - iv. fMRI
 - v. Pros and Cons of each method?
5. What are some of the important anatomical landmarks in the brain?
 - a. Basic Anatomy & Localization of Function
 - i. Lobes- Functions?
 - ii. Distributed Representation
 1. Structural Connectivity
 2. Functional Connectivity
 - a. Common Neural networks determined by resting-state fMRI
6. What are some of the important anatomical locations for some of our main cognitive functions?
 - a. Related to above question ^
 - b. Perception- receiving areas for the senses
 - c. Frontal Lobe
 - d. Broca's Area
 - e. Wernicke's Area
 - f. FFA

03 -PERCEPTION

1. What is the difference between Sensation and Perception?
 - a. Bottom-up versus top-down processing
2. Give some examples of how knowledge and previous experience play a role in forming our perceptions.
 - a. Where/How does knowledge and previous experience come in?
 - i. Object Recognition
 - ii. Speech Segmentation
 - iii. Pain Perception
3. Compare 4 theories of how we perceive/recognize objects
 - a. Helmholtz's unconscious inference
 - b. Gestalt principles of organization
 - i. Gestalt Laws- each one that we discussed
 - c. Considering Environmental Regularities
 - i. Physical regularities
 - ii. Semantic regularities
 - d. Bayesian Inference
 - e. Compare the 4
4. Explain plasticity in the perceptual system.
 - a. Vertical vs Horizontal lines
 - i. Selective Rearing
 - b. FFA- can you train FFA to respond to new stimuli?
 - i. Greebles
 - c. Somatosensory cortex
 - i. Homunculus
 - ii. Monkey studies
 - iii. Training/practice
 - iv. Phantom limb
5. What is the evidence from animal lesion studies and from Neuropsychology that there are 2 separate perceptual pathways? Compare and contrast the location and function of these pathways.
 - a. What vs Where pathways
 - i. What lobes?
 - ii. Monkey lesion study
 1. Object Discrimination
 2. Landmark discrimination
 3. Double dissociation
 - iii. Patient DF
 1. Agnosia
 2. Prosopagnosia
 3. Vs RV?

10 -VISUAL IMAGERY

1. Describe evidence for and against the argument that imagery and perception involve the same mechanisms.
 - a. Imagery and Perception
 - i. Spatial Correspondence
 1. Shepard & Metzler rotation tasks
 - a. Mental chronometry
 2. Mental Scanning task
 - a. Map Task
 3. Mental walks
 - ii. DEBATE- Spatial vs. Propositional
 - a. Pylyshyn- Not Spatial but propositional
 - b. Imagery works like perception- must be spatial
 - i. Viewing distance study
 - c. Interactions
 - i. Dim projection study
 - ii. Farah- Priming by imagery study
2. Describe similarities and differences in the brain mechanisms involved in imagery and perception
 - a. Kreiman et al
 - b. Le Bihan et al 1993
 - c. Ganis et al 2004
3. Describe how method of loci and Pegword techniques use imagery to improve memory.
 - a. Method of loci
 - b. Pegword technique

04- ATTENTION

1. Describe Inattentional Blindness and give some examples
 - a. Inattentional Blindness
 - b. Change Blindness
2. Compare and contrast the three main models of attention (Broadbent, Treisman, and MacKay).
 - a. Early
 - i. Broadbent's Early Selection Theory
 - ii. Cocktail Party Phenomenon
 - iii. Cherry's Shadowing Experiments
 - b. Attenuation
 - i. Treisman's Attenuation theory
 1. Dictionary Unit
 - c. Late
 - i. MacKay
3. Describe the Stroop task and the Attentional Blink task and their typical results. What do these tasks demonstrate about attention?
 - a. Stroop Task
 - b. Attentional Blink Paradigm

- c. Inattentional Deafness
- 4. Compare and contrast Overt and Covert Attention and give examples of each.
 - a. Overt vs Covert
 - i. Overt
 - 1. Eye-tracking
 - ii. Covert
 - 1. Helmholtz
 - 2. Precuing
- 5. Compare and contrast Automatic (Exogenous) and Voluntary (Endogenous) attention.
 - a. Voluntary
 - i. Divided attention
 - b. Reflexive
 - i. Saliency maps
 - ii. Physical vs meaning-based reflexive attention
- 6. What are some of the physical things and meaningful stimuli that can automatically affect the allocation of our attention?
 - a. Salience
 - i. Movement
 - b. Color
 - c. Meaningful stimuli
 - i. Faces
 - ii. Schemas
 - iii. Memory and Familiarity
 - iv. Addictive Stimuli

05 -STM & WM

- 1. Explain how information is processed in Atkinson and Shiffrin's modal model of memory.
 - a. Sensory Memory
 - b. STM
 - i. Control Processes?
 - c. Long-Term Memory
- 2. Describe evidence demonstrating the large capacity and rapid decay of sensory memory.
 - a. Sensory Memory
 - i. Echoic
 - 1. Duration?
 - ii. Iconic
 - 1. Capacity & Duration?
 - a. Partial Report task
- 3. Describe evidence demonstrating the limited capacity and relatively brief duration of short-term memory.
 - a. Duration?
 - i. Proactive Interference

- b. Capacity?
 - i. Miller says?
 - ii. But- visual/more complicated stimuli?
 - iii. Chunking
 - iv. Items or amount of info?
- 4. Contrast Baddeley's concept of working memory with Atkinson and Shiffrin's description of the short-term memory system.
 - a. WM versus STM?
- 5. Describe the functions of the components of Baddeley's revised working memory model.
 - a. WM- 3 parts (+ one added later!)
 - i. Phonological Loop
 - ii. Visuospatial sketch pad
 - iii. Central Executive
 - iv. Episodic Buffer
- 6. Describe evidence in humans and monkeys suggesting that the prefrontal cortex plays an important role in working memory.